Ex 2.2.1

Consider the context-free grammar:

S -> S S + | S S * | a

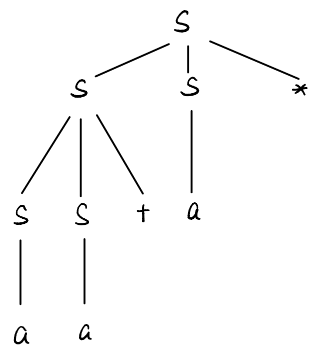
1. Show how the string `aa+a*` can be generated by this grammar.
2. Construct a parse tree for this string.
3. What language does this grammar generate? Justify your answer.

1) $S \Rightarrow SS*$

$\Rightarrow SS+S*$

$\Rightarrow aa+a*$

2)



3) L 为由 *, +, digits 组成的后缀表达式的集合.

证: 先证从 S 推导出的每个串都是后缀表达式.

对推导步数 n 进行归纳.

当 $n=1$ 时, 结果只可能是 a, 其在 L 中.

假设当 $n \leq k$ ($k \in \mathbb{N}_+$) 时, 结果均在 L 中,

则当 $n=k+1$ 时,

$$S \underset{lm}{\Rightarrow} SS+ \overset{*}{\underset{lm}{\Rightarrow}} xS+ \overset{*}{\underset{lm}{\Rightarrow}} xy+$$

或 $$S \underset{lm}{\Rightarrow} SS* \overset{*}{\underset{lm}{\Rightarrow}} xS* \overset{*}{\underset{lm}{\Rightarrow}} xy*$$

由于从 S 到 x, y 的推导至多为 k 步, 故

x, y 为后缀表达式 (i.e., 在 L 中), 因此

$xy+$ 和 $xy*$ 亦为后缀表达式, 在 L 中.

下证每个后缀表达式均可由 S 生成.

对表达式的长度 n 进行归纳 (注意到 n 为奇数)

当 $n=1$ 时, 其只能是 a, 可由 S 生成.

假设当 $n \leq 2k-1$ ($k \in \mathbb{N}_+$) 时,

后缀表达式可由 S 生成. 则当 $n=2k+1$ 时,

设后缀表达式为 w, 其形式必为

$$x\ y\ + \text{ 或 } x\ y\ *$$

其中 x, y 为长度不超过 $2k-1$ 的串, 可由 S 生成.

故 $$S \underset{lm}{\Rightarrow} SS+ \overset{*}{\underset{lm}{\Rightarrow}} xS+ \overset{*}{\underset{lm}{\Rightarrow}} xy+$$

$$S \underset{lm}{\Rightarrow} SS* \overset{*}{\underset{lm}{\Rightarrow}} xS* \overset{*}{\underset{lm}{\Rightarrow}} xy*$$

即 $w = xy+$ 或 $xy*$ 可由 S 生成.

□

Ex 4.2.1

Consider the context-free grammar:

```
S -> S S + | S S * | a
```

and the string aa + a*.

1. Give a leftmost derivation for the string.
2. Give a rightmost derivation for the string.
3. Give a parse tree for the string.
4. ! Is the grammar ambiguous or unambiguous? Justify your answer.
5. ! Describe the language generated by this grammar.

1) $S \underset{lm}{\Rightarrow} SS* \underset{lm}{\Rightarrow} SS+S* \overset{+}{\underset{lm}{\Rightarrow}} aa+a*$

2) $S \underset{rm}{\Rightarrow} SS* \underset{rm}{\Rightarrow} Sa* \underset{rm}{\Rightarrow} SS+a* \overset{+}{\underset{rm}{\Rightarrow}} aa+a*$

3) 同 Ex 2.2.1 2)

4) 无二义性. 任意一个句子只有一个最左和最右推导.

5) 同 Ex 2.2.1 3)

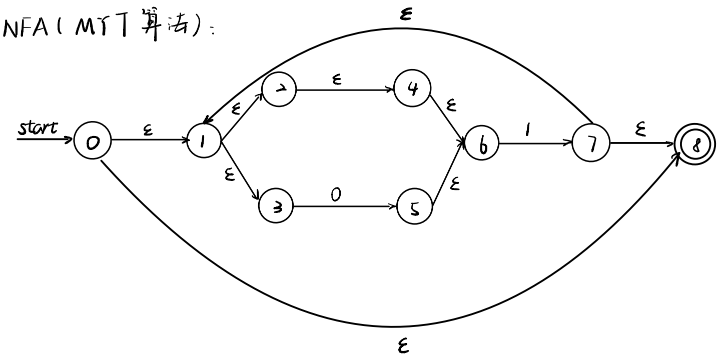
Ex 4.2.3

Design grammars for the following languages:

1. The set of all strings of 0s and 1s such that every 0 is immediately followed by at least one 1.
2. ! The set of all strings of 0s and 1s that are palindromes; that is, the string reads the same backward as forward.
3. ! The set of all strings of 0s and 1s with an equal number of 0s and 1s.
4. !! The set of all strings of 0s and 1s with an unequal number of 0s and 1s.
5. ! The set of all strings of 0s and as in which 011 does not appear as a substring.
6. !! The set of all strings of 0s and 1s of the form xy, where x<>y and x and y are of the same length.

1) 正则表达式：$((\varepsilon|0)1)^*$

NFA(MYT算法)：



$A_0 \longrightarrow A_1 \mid A_8$

$A_1 \longrightarrow A_2 \mid A_3$

$A_2 \longrightarrow A_4$

$A_3 \longrightarrow 0A_5$

$A_4 \longrightarrow A_6$

$A_5 \longrightarrow A_6$

$A_6 \longrightarrow 1A_7$

$A_7 \longrightarrow A_1 \mid A_8$

$A_8 \longrightarrow \varepsilon$

$S \rightarrow 1S \mid 01S \mid \varepsilon$

2) $S \longrightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \varepsilon$

3) $S \longrightarrow 0S1S \mid 1S0S \mid \varepsilon$ 或 $S \rightarrow 0S1 \mid 1S0 \mid SS \mid \varepsilon$

4) $S \longrightarrow S_0 \mid S_1$

$S_0 \longrightarrow 0 \mid 0S_0 \mid S_0 0 \mid 1S_0S_0 \mid S_0 1 S_0 \mid S_0 S_0 1$

$S_1 \longrightarrow 1 \mid 1S_1 \mid S_1 1 \mid 0S_1S_1 \mid S_1 0 S_1 \mid S_1 S_1 0$

5) $S \longrightarrow 1S \mid 0S_1 \mid \varepsilon$

$S_1 \longrightarrow 0S_1 \mid 1S_2 \mid \varepsilon$

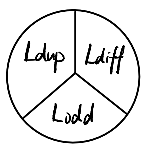
$S_2 \longrightarrow 0S \mid \varepsilon$

6) $L_{dup} = \{ww \mid w \in \Sigma^*\}$

$L_{diff} = \{xy \mid x \neq y \land x \in \Sigma^* \land y \in \Sigma^*\}$

$L_{odd} = \{w \mid |w| \text{ is odd} \land w \in \Sigma^*\}$

$\Rightarrow$



Σ^*.

L_{odd} 上下文无关 $\Rightarrow$ L_{diff} 上下文无关

从L_{diff}中取出长度相等的xy组成的串，$x \neq y$.

设 $x_k \neq y_k$.

$k \quad n-1-k$

则 $xy = \underbrace{x_0x_1\cdots}_{k} x_k \underbrace{x_{k+1}\cdots x_{n-1}y_0y_1\cdots}_{n-1} y_k \underbrace{y_{k+1}\cdots y_{n-1}}_{n-1-k}$

$= \Sigma^k x_k \Sigma^k \Sigma^{n-1-k} y_k \Sigma^{n-1-k}$

在Σ上长为k的串

k和n取值任意，故

$W \longrightarrow 0 \mid 1$

$A \longrightarrow 0 \mid WAW$

$B \longrightarrow 1 \mid WBW$

$S \longrightarrow AB \mid BA$